

MSc Dissertation · Trinity College Dublin

Only the Eye of the Bird

Adaptive Visual Noise Cancellation for Attention Guidance
on Consumer Passthrough Headsets

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School of Computer Science and Statistics · Meta Quest 3, Unity 6, one shader, nine filters · 11 participants, 4 sessions

Before the argument — here is the thing itself.



01 · THE WINDOW

Full-quality reality

The app draws nothing inside it. Both eyes, 110°, no added lag.

02 · THE FILTER

Processed edges

One shader fades the world outside. Nine built, two tested.

03 · THE PASS-THROUGH

Signals still get through

A detector opens a hole over traffic lights and signs.

The argument in five moves.

01

The idea

Two senses answer distraction by subtraction. Sight has none.

02

The walls

You may draw in front of reality, and nothing else.

03

The system

Draw nothing where it matters. One shader, nine filters, three adaptations.

04

The study

Eleven people. Real hardware and simulation. Benefit and price together.

05

What we found

Focus up.
Awareness intact.
The edge of vision got better.

01

PART 01

The idea

A children's film, a library carrel and a pair of headphones · then roughly 110 papers that pruned the design before a line of it was written.



THE ORIGIN

“I can only see the eye of the bird.”

A STORY ABOUT ATTENTION

The scene

Droṇa sets his students one test: hit the eye of a wooden sparrow in a distant tree. Before each shot he asks what they can see.

One says the tree. One says the branches, the leaves, the sky, his brothers. Arjuna says: **“only the eye of the bird.”** He is the only one told to shoot.

WHY IT STUCK

Not the archery — the absorption

A person can be so far inside a task that everything else stops arriving. I have had sparks of it — fifteen minutes, maybe twenty, and you cannot summon them back.



SO THE QUESTION

If that state is everything else falling away, **can a headset make some of it fall away for you?**

Two of our three senses already have an answer. Sight has none.

SPACE

The library carrel

Nothing is added. The room is subtracted until only the page is left.

SOUND

Noise-cancelling headphones

Not louder lectures. Everything else quieter.

SIGHT

...nothing

Arrows, labels, badges. Every visual aid on the market **adds**.

THE NAME FOR IT

Visual noise cancellation

camera → processing → display is exactly
microphone → processing → speaker.

THE PROPOSAL

Headphones for your eyes

Mark the part of the world you are working on.
Everything outside it fades.

WHY ONLY NOW

Every pixel is already synthetic

Cameras capture the room and the headset rebuilds it. So it can be changed.

BUT AN INTUITION IS NOT A FINDING · Three things: the clutter must cost something, removing it must give that back, and it must run on hardware people already own.

The cost is real — and it shows up in mistakes, not in slowness.

CLAIM 1, ANSWERED

Peripheral clutter · 66 people

2.4× more wrong presses (1.33 → 3.15) and **8× more misses** (0.14 → 1.18).

Speed did not change at all. **Mistakes are the currency.**

MECHANISM

Irrelevant sights spend the same
headspace

Processing you did not need competes for the
pool the task needs.

WHY DIMMING

Attention chases contrast

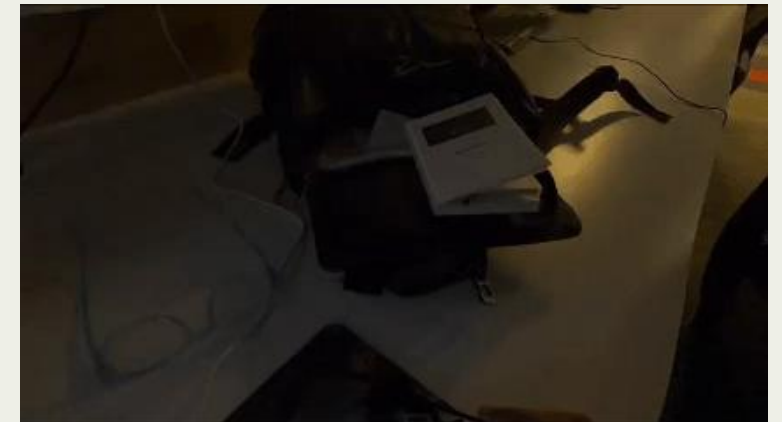
Flatten it and the periphery stops shouting.
Strength is a dial, not a switch.

CLAIM 2, PARTLY ANSWERED

Someone had already dimmed distractions

Engineers and NASA staff, assembly under distraction. Dimming
everything irrelevant beat dimming only the offender.

The blunt instrument won — remember that at the smart filter.



The literature didn't just motivate the design. It pruned it.

Three of the four most obvious ways to build a visual filter are **contraindicated by published evidence**.

What the evidence said	Where it comes from	What we therefore decided
Whole-periphery blur was the most noticed of five guidance techniques — 59 of 102 people reported it, against 12 for a red dot and 3 for a zoom.	Grogorick, Albuquerque & Magnor (2018), N=102 — post-experiment subtleness questionnaire	Blur is built, but never carried into testing.
Draining colour barely steers attention. Darkening an area does.	Wang, Gan & Li (2020) — eye-tracked, head to head in 360° video	Greying is only ever an ingredient , never a filter of its own.
Vignetting tied to head movement increased sickness in 11 of 15 participants, who also preferred the control condition without it.	Norouzi, Bruder & Welch (2018), N=15, velocity- and acceleration-based vignetting	Invert it — the effect yields while you move and returns when you settle.
People choose partial fading over total removal — and anxiety rose in every narrowed field.	Cheng et al., CHI (2022) for the preference; Barhorst-Cates et al. (2016), 116 people across four experiments, for the anxiety	Blackout is the extreme of the range , not the default.
No manipulation of this kind goes unnoticed.	Sutton et al., UIST (2022) — the closest predecessor to this work	Stop chasing invisibility ; design for noticeable but tolerable.
Errors move under distraction. Reaction time does not.	Ai et al. (2025), N=66 — 2.4× wrong presses, 8× misses, speed unchanged	Accuracy is the primary measure , fixed before data collection.

Each row maps a specific published result onto a specific design decision. The first four are the rejections; the fifth set the design target and the sixth set the outcome measure.

02

PART 02

The walls

On the hardware people actually own, the obvious way to build this is not difficult. It is forbidden, three times over.



The room reaches you as a camera feed — and you may only draw in front of it.

The display — what the user sees
assembled by the operating system

Your application's layer
full control — but only drawn IN FRONT

▲ the system stacks these together ▲

The rebuilt view of the real room
cannot read it · cannot change it · cannot remove it

AND THE WALL IS DELIBERATE

Privacy and safety. Apple is stricter still — those pixels need a licence that cannot ship in the consumer store.

SO WE HAD TO GET CREATIVE · Nothing **local** is possible at Level 1, and Level 3 gives back a worse view than the one you replaced. Which leaves Level 2: draw over the world, and leave a hole where you want the truth.

LEVEL 1 · RECOLOUR THE WHOLE VIEW

Free, and useless here

Whole-room brightness and colour, no camera pixel needed — but **uniformly, or not at all.**

LEVEL 2 · DRAW OVER IT

What this system is built on

Your own shapes in front: dim, black out, **or leave a hole.** Darken or hide only.

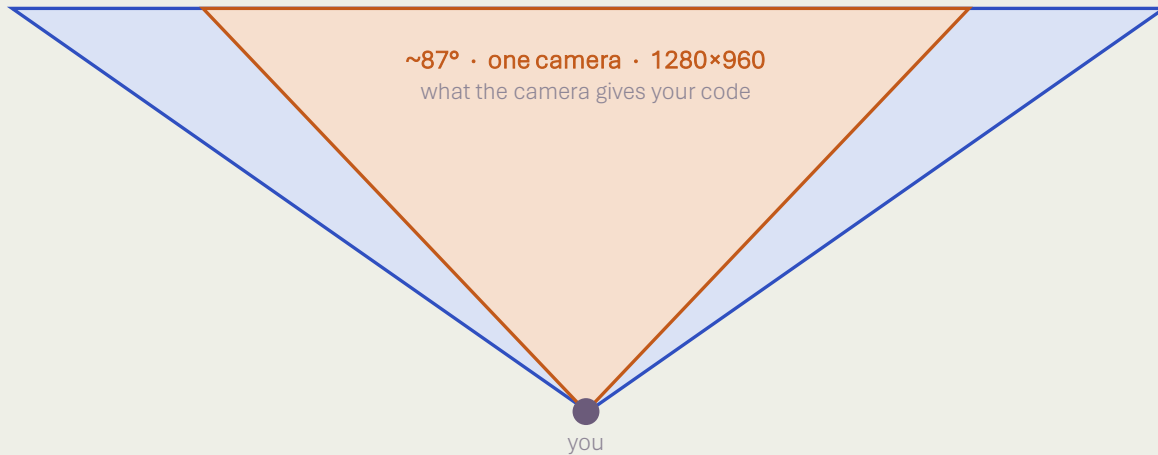
LEVEL 3 · TAKE THE CAMERA FEED

Anything, pixel by pixel — at a price

Raw camera, but **one camera, 1280×960, ~87°** against the ~110° stereo view you had.

The camera you are given sees less than the view it would replace.

~110° · both eyes · higher resolution · no added delay
what the headset already shows you



THE TRAP

The straightforward version backfires

Paint the processed feed over everything and you hand back one camera, lower resolution, extra lag, and a view **23° narrower on each side** — losing the far edge.

AND A LIMIT ON WHAT YOU CAN TEST

Already below normal eyesight

No camera-based headset matches human acuity.

So: how do you process the edges without degrading the middle?

03

PART 03

The system

One idea does most of the work, and it is close to a trick: the best way to show someone reality is to draw nothing at all.



The best way to show someone reality is to draw nothing at all.

THE CORE MOVE

Inside the window, the app draws nothing

Not “transparent” — **nothing**. So what reaches the eye is the headset’s own view: full quality, full 110°, free.

AND IT ANSWERS THE CAMERA WALL

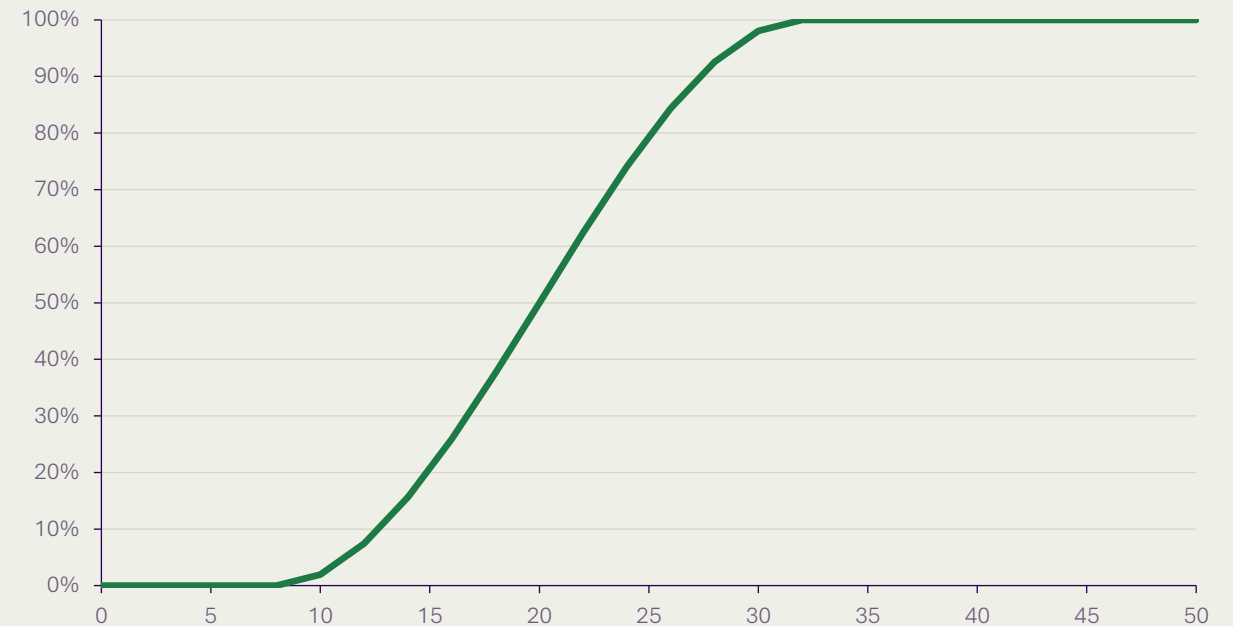
Everything is spent at the edges

All the work happens where eyesight is worst, so the narrow camera stops mattering.

And it appears to be new.

WHAT THE CHART SHOWS

How hard the filter works, by angle from the centre of the window



angle away from the centre of the window (degrees) — clear to 8°, half strength at 20°, full from 32°

The window has to belong to the room, not to your head.

WRONG · WHAT YOU GET FOR FREE

Stuck to your head

Screen coordinates follow your gaze like a torch beam — and head-coupled restriction makes people sick.

BETTER · THE DEFAULT

Locked to a direction

A bearing and elevation. Stays over the desk when you turn — but not when you walk.

BEST · WHAT THE STUDY USED

Anchored to the room's real geometry

Four corner rays on release, remembered as physical points. Walk toward it, around it, away.

[VIDEO / GIF]

Walking around an anchored window

Anchor to an object, walk toward it, around it, back. Shot on device.

ENGINEERING NOTES

No shader change — it already consumes a bearing-and-elevation rectangle. Fails softly back to direction-locking if a corner ray misses.

Correct in each eye. Completely broken when you use both.

THE BUG

Sampled by screen position

Your two eyes see the same point at **different places on their screens** — so each was handed a different camera pixel for the same object. Ghosting.

THE FIX

Sampled by world direction

Ask which **direction** the pixel is looking instead. A direction is the same for both eyes — so it merges, like paint on the walls.

THE TRANSFERABLE BIT · Per eye, the original code was correct. It failed only as a **binocular system** — on a device, with a person inside it.

Good noise cancelling doesn't mute everything. It lets the announcement through.

THE ANALOGY, RETURNING

What headphones actually do now

Modern ANC passes the announcement through and suppresses the rest. Nobody calls that a compromise.

THE CLAIM

So the visual version cannot be a constant

Dim the street, but let the traffic light through. Deciding what passes **is** the design.



WHAT THAT RULES OUT

A constant is a blindfold

Always on, everywhere, at full strength is not noise cancellation.

Three things the filter adapts to.

01 · WHAT IS IN THE SCENE

A detector opens holes for real signals

Boxes for traffic lights and signs. The filter lifts off each one, pushes it forward, and darkens a ring around it.

02 · HOW BRIGHT IT IS

Glare is rolled off, never clipped

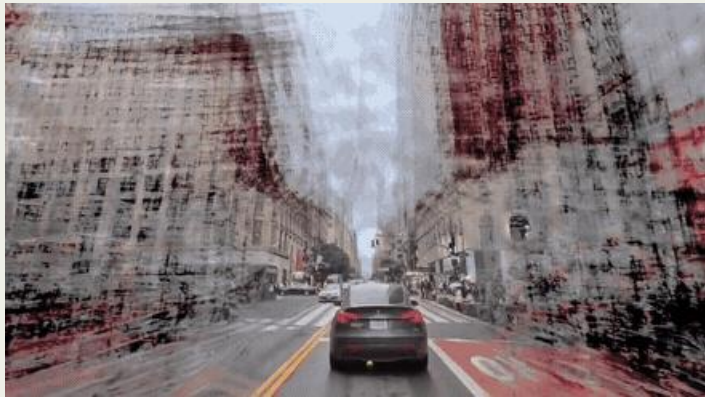
Bright colourless pixels compress instead of clipping. A ring test exempts real lamps.

03 · WHAT YOU ARE DOING

The filter yields while you move

Out in 0.20 s when you turn, back over 0.70 s once you settle.

Nine ways to fade the edges of the world. One shader.



ONE SHADER, ONE GEOMETRY • A filter is one rule for colour and opacity by distance outside the window — which is why nine cost almost nothing to maintain. ★ **Hard Dark** and **SignPop** are the two carried into testing.

Each cell is a short looping clip of that filter running on device.

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Each cell is a short looping clip of that filter running on device.

Two were rejected because the literature had already ruled them out.

BLUR — THE ORIGINAL FLAGSHIP

Peripheral blur announces itself

Detected **through the contrast collapse it causes** — and the most noticed of five techniques with 102 people.

Also the heaviest shader path — 17 taps per camera.

AND ONE NEVER PROMOTED TO A FILTER AT ALL

Greying out

Head to head with eye tracking, darkening steered attention and colour-draining did **almost nothing**. It survives only inside ColorPop.

CHROMATIC COOL

The losing arm of its own source study

In the study it came from, **luminance beat chroma**. The mode implements the weaker arm.



Three more were rejected because we watched people use them.

CONSPICUITY SQUEEZE

It brightened the periphery

At the settings piloted it did the opposite of what it should. Confirmed in the shader.

OUTLINED DARK

It was more interesting than the task

The surviving edges read as a **comic-book style** testers preferred to the task.

GRANULATED

Lost an informal head-to-head

Less subtle than SignPop, and grains occlude **detail** — the wrong failure mode for driving.

ONE COULD NOT BE BUILT; ONE THERE WAS NO ROOM FOR

SPOT LIFT would brighten the window rather than darken the rest — but the OS passthrough layer cannot be brightened, so it never left the video path.

SOFT DARK was built, but a confirmatory block does not fit a ~53-minute session.

THE HONEST WORDING

Two of the three are **settings-dependent, not technique-dependent**, and neither parameter was swept. So: “**rejected at the settings piloted**”.

Two filters went into testing — the strongest and the smartest.

SURVIVED · THE REAL-HARDWARE ARM

Hard Dark

Total blackout outside the window — **the maximum of the range**, and pure luminance change, so nothing but brightness differs.



SURVIVED · THE SIMULATION ARM

SignPop — built for this work

A composite, not a variant. ColorPop's grading gated to detected signals, plus a highlight with a darkened surround and a ~ 1 Hz pulse. No paper proposes it.



NINE BUILT, TWO EVALUATED, FOUR REPORTED AS DESIGN POINTS · The tiers are not equally evidenced, so this deck names which is which.

04

PART 04

The study

One test on real hardware, one in simulation, the same eleven people, and a reframing of what is actually being claimed.



This does not make you focus better. It makes distraction cost you less.

✗ NOT THE CLAIM

“A filter that makes you more attentive.”

Dimming adds no capacity and trains no skill. In a quiet room it offers you nothing.

✓ THE CLAIM

“A filter that stops something else spending it.”

Suppress what steals headroom and you get it back. The benefit needs distraction to exist.

QUESTION 1 · REAL HARDWARE

Test 1 · the benefit

Can you do **less badly** under distraction with the filter? Hard Dark, anchored to a desk.

QUESTION 2 · SIMULATION

Test 2 · the price

Does the smart version work **without costing awareness**? Benefit and price, same task, same moment.

WHY ONE OF EACH · Live passthrough can never repeat the same peripheral events; the video sphere can, with machine-timed ground truth. But a simulation is not the real thing.

Each hypothesis stated with the observation that would refute it.

	What we predicted	What would have refuted it	Outcome
H1	Accuracy on a demanding task with distractions present is higher with the filter than without .	No difference, or a difference favouring no-filter.	Supported — +3.65 points, 9 of 11
H2	Awareness of peripheral events is not meaningfully worse with the filter — specifically, not more than 10 points worse.	A drop larger than 10 points, which would be a conclusive finding that the system harms situational awareness.	Supported on the safety side — a loss over ~5 points is ruled out
Secondary	If there is an effect on peripheral detection, it should track where the filter actually acts .	A flat profile across all angles, or a peak where the filter does nothing.	Consistent — the gain sits in the 20–30° ring

ON H2

Why a limit was set in advance

“We found no difference” is weak. Naming the smallest loss you would care about — **10 points** — before collecting turns it into a claim that could have failed. It did not.

The 10-point margin is a pre-registered judgement, not a value taken from any paper: no study in this literature supplies one. It was set from the focus-versus-awareness trade-off that the diminished-reality work documents repeatedly, and fixed before collection. Lakens (2017) supplies the two-one-sided-tests procedure and is explicit that the bound must come from the researcher’s smallest effect size of interest.

A memory task in a real room. A detection task on a moving scene.

REAL HARDWARE · HARD DARK

What the participant does

Shapes one at a time; press when one matches **the one immediately before**. It holds and refreshes memory on every trial.

[VIDEO / GIF]

Test 1, without then with the filter

SIMULATION · SIGNPOP

What the participant does

Ring markers at known angles on a moving scene; press when you notice one. Onset and catch are machine-timed.



A CHECK DONE BEFORE THE ANALYSIS · Markers are drawn **through** the filter, not on top — so filtered was harder to see, not easier.

WHAT WAS COLLECTED

Twelve people over four days. Eleven usable pairs per task.

Phase	People	What changed afterwards
Pilot · day 1	5	The memory task was too easy — nearly everyone scored close to perfect, leaving no room to show an improvement. Made harder for later sessions.
Extra · days 2–4	7	Harder task, larger pool of distraction material.

Both phases are marked separately on every chart — **two independent samples agreeing** is this study’s nearest thing to a replication.

CEILING EFFECT

Why “too easy” is a problem

Near 100% without the filter leaves nothing to improve, so day-1 **understates** the effect.

METHOD

How a catch was scored

A press counts within **10°** of a live marker — scored against where the marker actually was at that instant, not an average.

Participant numbers only; no names appear anywhere in the analysis record. One participant’s run on the moving-scene task is excluded on documented evidence of non-responsive triggering.

05

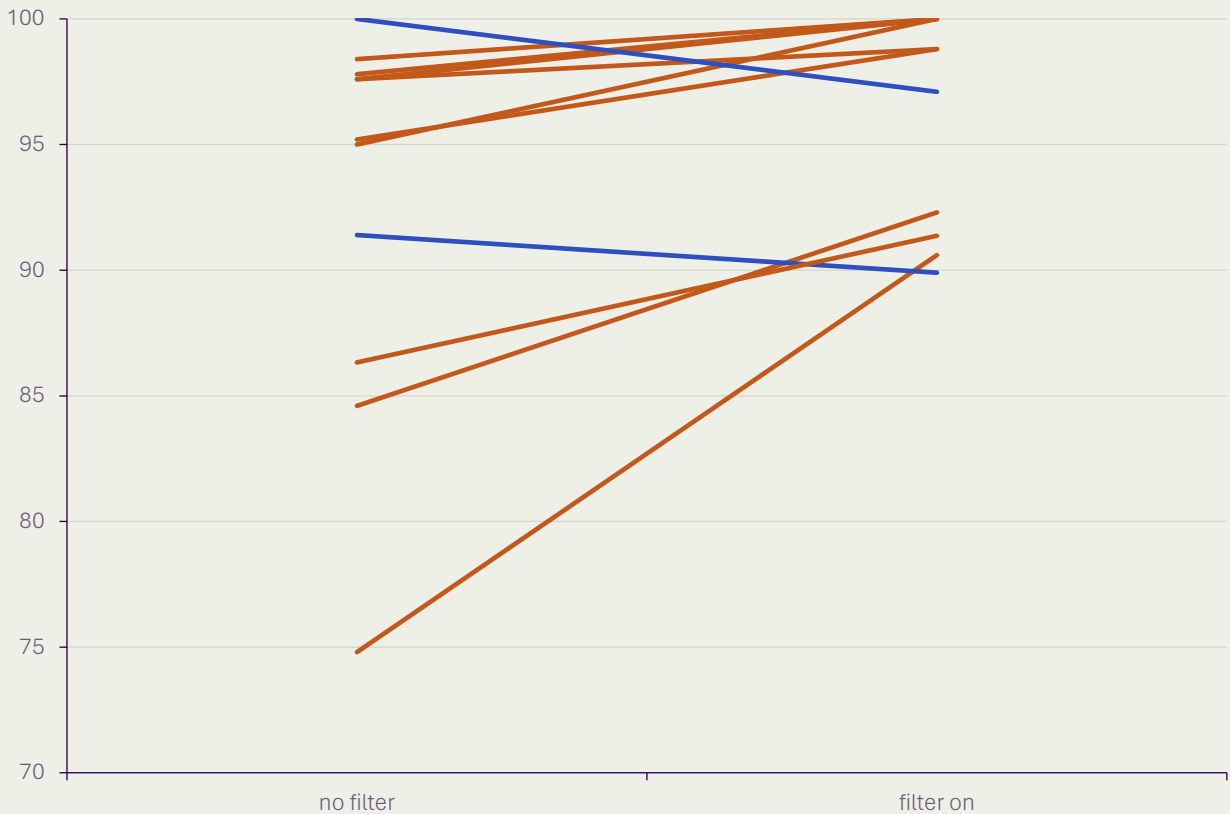
PART 05

What we found

Focus up, awareness intact, and one result that needed a great deal of checking before I believed it.



Accuracy up 3.65 points, and it held for nine of eleven people.



92.6%

no filter

96.3%

filter on

+3.65 points

CI +0.28 to +7.02 · p = 0.036 · Wilcoxon p = 0.024 · 9 of 11 improved.

Phase	off	on	gain	improved
Pilot	97.2	99.4	+2.20	4/4
Extra	90.0	94.5	+4.48	5/7

The pilot’s smaller gain is the ceiling: at 97.2% only 2.8 points were available. **A harder task doubled the benefit.**

Awareness of the surroundings was not lost. If anything it went up.

+3.9

points

more markers noticed with the filter on, over the whole field of view

THE SAFETY QUESTION, ANSWERED

A loss beyond about 5 points is ruled out

The line was fixed at 10 points before collecting. Range -4.9 to +12.6 — worst case supported is about **five points lost**.

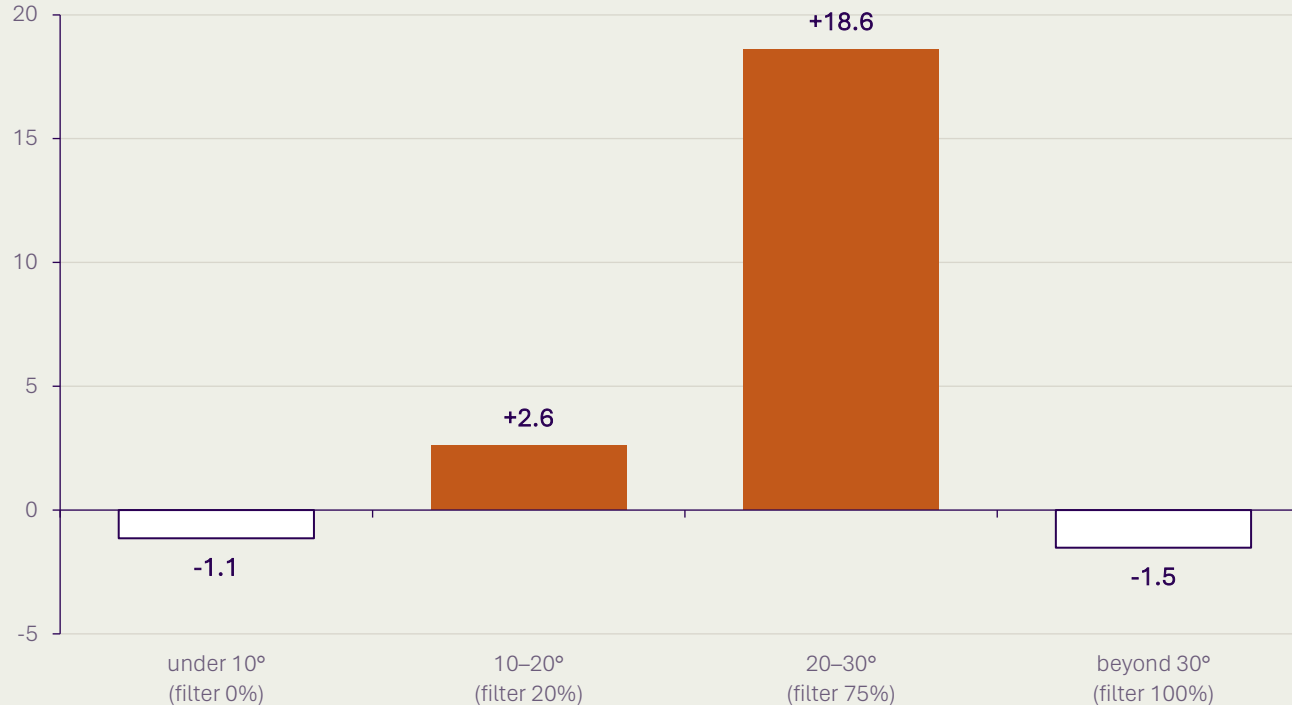
WHY THIS IS THE RESULT THAT MATTERS MOST

A focus aid that works by blinding you is a hazard, not a focus aid

Every study documents the same trade-off. **This is the first measurement of it on real consumer hardware** — and it comes out better than the literature would predict.

ONE THING THE POOLED NUMBER CANNOT DO · It mixes regions the filter barely touches with regions it owns completely. **Where it actually did something is the next slide.**

Better exactly where the filter acts and the eye still works.



A METHOD POINT THAT CHANGED THE ANSWER

Markers travel ~17° while visible and 45 of 82 cross a band, so every press was scored against the marker's actual position at that instant.

+18.6 points

40.0% → 58.2%

CI +2.5 to +34.7 · $p = 0.028$ · Wilcoxon $p = 0.030$ · **9 of 11 improved.**

THE SHAPE IS THE FINDING

Both conditions are needed

Below 10° there is no filter to lift. Beyond 30° the eye cannot use it. **Only 20–30° meets both** — as the shader's geometry predicts.

DOES IT REPEAT?

Pilot **+25.4** (4/4) · Extra **+14.7** (5/7) — the only band agreeing across phases. Brief markers: 2/36 → 9/30.

What the numbers mean, and how much weight they carry.

THE p-VALUE

How often chance alone would do this

If the filter did nothing, how often would a gap this big appear anyway? **p = 0.036** — about one run in twenty-eight.

THE 95% CONFIDENCE INTERVAL

The range the data supports

+0.28 to +7.02. It excludes zero, so the effect is real. It is 6.7 wide, so its size is not.

THE WILCOXON W

The test that assumes nothing

Scores near 100% break a t-test’s assumptions. Wilcoxon ranks each person’s own change. **W = 8.0, p = 0.024.**

EFFECT SIZE AND POWER

How big, and how likely we were to see it

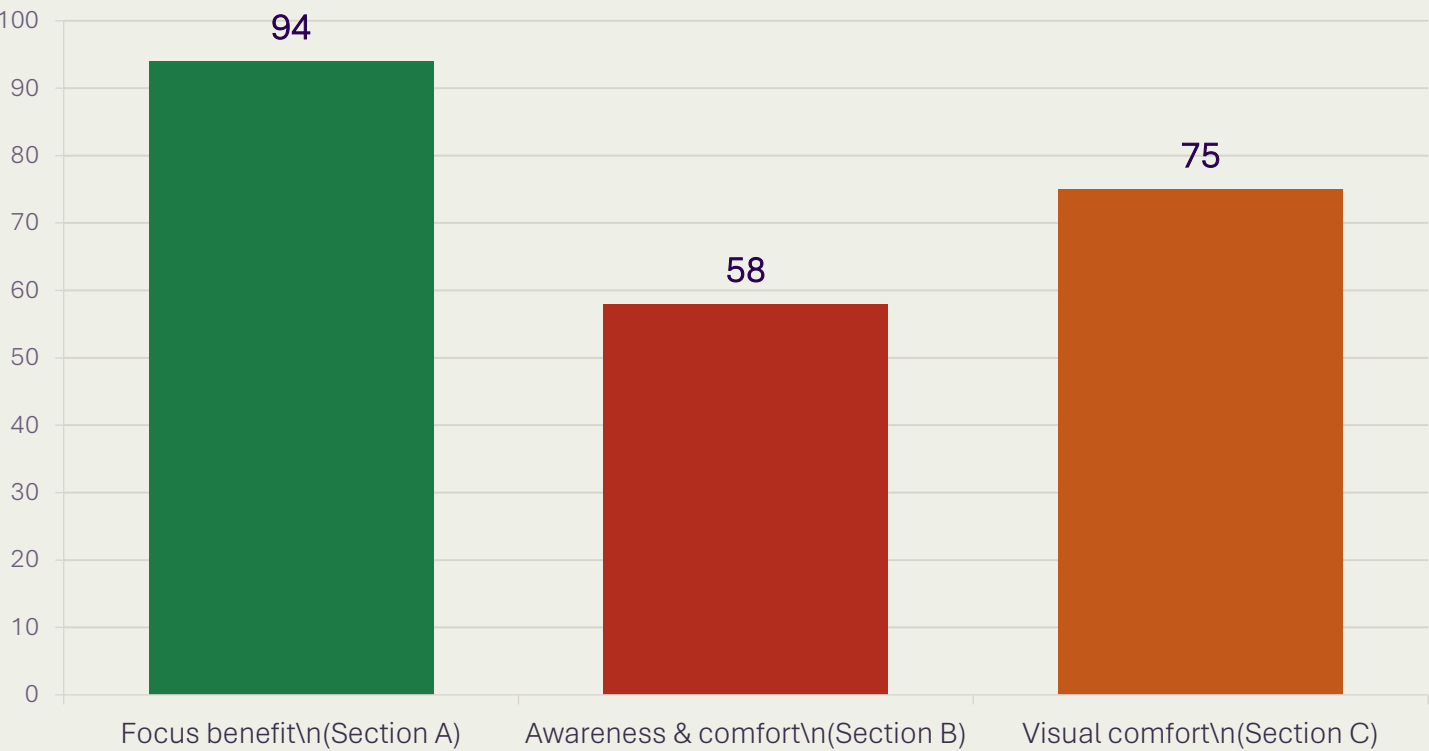
dz = 0.73, medium-to-large. **Power 0.59** — about 17 people would reach the conventional 80%.

Result	Change	95% CI	t (df 10)	p	Wilcoxon	dz	Power (n for 80%)
H1 · memory task, Hard Dark	+3.65 pp	+0.28 to +7.02	2.42	.036	W = 8.0, p = .024	0.73	.59 (17)
H2 · awareness, whole field	+3.9 pp	−4.9 to +12.6	—	TOST vs ±10 pp	—	—	—
Secondary · the 20–30° ring	+18.6 pp	+2.5 to +34.7	2.57	.028	W = 9.0, p = .030	0.78	.64 (16)

WHAT IT ADDS UP TO · Both headline results clear the threshold on **two tests with different assumptions**. What eleven people cannot buy is a narrow interval.

Paired t-tests on per-participant condition means, with Wilcoxon signed-rank as the assumption-free confirmation; dz is the within-person standardised effect size; power computed at the observed effect for n = 11. H2 uses two one-sided tests against the pre-registered 10-point margin (Lakens, 2017).

What the ten participants said, in one picture.



share of answers that came out in the filter’s favour — 47/50, 29/50 and 30/40 items across ten participants

WHAT THE SCORE IS

A count, not a rating

The share of answers favouring the filter, with the reverse-worded items flipped back.

THE SHAPE OF IT

Clear focus benefit, middling comfort, and a **noticeably softer awareness score.**

Section D was not administered; one comfort item was answered by nobody.

PARKED FOR NOW · What people **felt** against what was measured belongs in the discussion, not here.

Ten post-session questionnaires, scored from the blank template’s answer key; reverse-keyed items flipped before scoring. Section A five items, Section B five items, Section C four scored items.

It was not the dimming. It was dimming plus letting the signal through.

WHAT THE CODE AND DATA SAY

Nearly every marker sits on a detection

70 of the 80 targets (88%) fall inside an active detection window. In the no-filter arm the detector is suppressed.

SO THE MECHANISM IS CONCRETE

The pass-through **is** the effect

The marker stays dimmed; the object beneath it pops. People were spotting **the traffic light**, not a brighter marker.

THE PRICE OF THAT CLARITY

The manipulation is a bundle

Filter-on switches everything at once — dimming, detection, the hole, the boost, the surround, the pulse. **No number splits the +18.6.**



COVERAGE MEASURED, NOT ASSUMED · Against the baked detection track, replicating the runtime's own gates — minimum lifetime 0.6 s, minimum box size 1°, 16-window cap.

Built, found, and not claimed.

C1 – C2

Built

A subtractive interface on a headset you can buy. Nine filters in one shader, on a platform that forbids touching reality at all.

The two closest published studies had to simulate this inside VR.

C3

Found

Focus +3.65 points, 9 of 11.

Awareness not lost — beyond ~5 points ruled out.

Mechanism +18.6 in the ring the filter aims at.

THE HONEST BOUNDARY

Not claimed

The manipulation is a bundle. Eleven people, not twenty.

Half the participants **felt** cut off and nothing measures that. No claims about driving.

WHAT COMES NEXT

1 More participants — about 17 reaches the conventional 80% and tightens every interval. **2 More sessions, more targets in the thin bands** — so the ring result rests on a confirmatory test. **3 Measure the feeling** — audio or haptic cueing leaves vision untouched.

Real cost, recoverable, and buildable on a headset you can buy. A filter that dims the surroundings but lets the important things through bought a real gain in focus, cost no measurable awareness — and made the edge of vision better, not worse.

ONLY THE EYE OF THE BIRD

Thank you.

A filter that dims the surroundings but lets the important things through — **and, in the ring where it acts, made the edge of vision better, not worse.**

Shreyansh Soni Supervisor: Dr. John Dingliana

School of Computer Science and Statistics · Trinity College Dublin

Questions welcome — any of the clips in this deck can be replayed.